

A Quantitative Survey of Motion Forecasting in Autonomous Driving

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Abstract—Anticipating the future trajectories of vehicles, pedestrians, and other traffic agents is essential for safe autonomous driving. Motion forecasting has evolved rapidly, driven by large-scale benchmarks, probabilistic evaluation metrics and advances in model architectures, from recurrent and graph-based networks to Transformers and, more recently, large language models (LLMs). This rapid growth has made it difficult to systematically track general trends. This survey addresses this gap through a statistical, data-driven review of motion forecasting for autonomous driving published in recent years. We quantify the evolution of the field along four axes: input representations and features; the prediction scope, contrasting single-agent and joint multi-agent forecasting; datasets and benchmarks adopted and the prediction metrics used to report performance. Based on this analysis, we organize the surveyed models into an architectural taxonomy. We conclude by discussing open challenges and research gaps that emerge directly from these trends, offering a quantitative reference point for researchers.

I. INTRODUCTION

Accurately anticipating the future motion of vehicles is a key capability of autonomous driving systems. Motion forecasting models consume the past and present state of traffic agents and their surroundings to estimate plausible future trajectories, providing the advance information that planning and decision-making modules rely on to act safely. Over the past several years, this task has attracted intense research interest, powered by the release of large-scale, highly annotated driving datasets and rapid advances in deep learning architectures.

The modeling landscape underlying this progress covers several paradigms, from physics-based and rule-based formulations to increasingly capable learned architectures, each reflecting different assumptions about how future motion should be reasoned about. Graph-based and Transformer-based encoders have coexisted since the earliest works considered in this survey, such as *PAGA* [18] and *Scene Transformer* [62], with recurrent units such as GRUs and LSTMs persisting mainly as embedded sequential components within these designs, for example to encode agent history in *FJMP* [74], rather than as standalone forecasting backbones. Attention-based Transformer encoders soon consolidated as the dominant architectural choice, exemplified by *Wayformer* [61], *QC-Net* [117] and *HDGT* [35], establishing a foundation that much of the subsequent literature builds upon. From this foundation, the field has diversified into several new sub-paradigms. An autoregressive, GPT-style formulation of sequence generation, as in *MotionLM* [76] or *DriveGPT* [31]; diffusion generative modeling, from *MotionDiffuser* [36] to *TrajFlow* [101] and, most recently, the repurposing of pretrained large language

models themselves as the forecasting or reasoning backbone, as *CoT-Drive* [50]. Physics and rule-based reasoning, meanwhile, has not been replaced by this progression, it persists throughout the surveyed period, from early adversarial robustness studies grounded in vehicle kinematics [4] to a renewed interest in combining learned prediction with explicit kinematic constraints and driving-style priors [91], [13].

Motion forecasting methods can be broadly distinguished by the type of input they rely on. Most approaches follow a *scene-based* paradigm, operating on structured, vectorized representations of the scene, including agent positions, velocities, headings and map elements using HDMaps. In contrast, *raw-perception-based* works operate directly on raw sensor inputs, consuming camera images [2], LiDAR point clouds and other sensor data within the forecasting pipeline itself to bypass intermediate structured representations. While the two families address the same underlying task, they differ substantially in their input processing and core design assumptions.

As illustrated in Fig. 1, the analysed corpus highlights a sustained increase in motion forecasting research since 2022. Notably, scene-based structured representations predominate in modern architectures, while end-to-end raw perception approaches remain a minority.

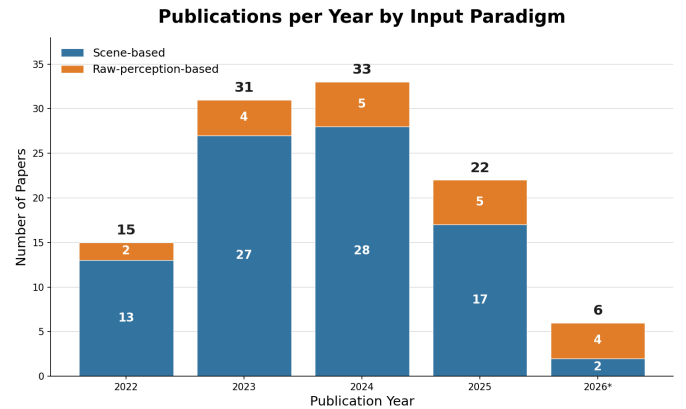


Fig. 1. Annual distribution of the surveyed papers (2022–2026), categorized by input paradigms. The 2026 count (marked with *) is partial due to ongoing data collection.

This survey aims to provide a clear explanation of the most recent articles on trajectory prediction from a statistical perspective, covering important aspects such as datasets, metrics and the most commonly used methods in recent times. The remainder of this paper is organized as follows. First, Section II formalizes the motion forecasting problem. Subsequently, Section III reviews the most widely used public benchmarks, while Section IV details the prediction metrics adopted to evaluate model performance. Following this, Sec-

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tion V organizes the surveyed models into an architectural taxonomy. Finally, Section VI outlines open challenges and future research directions emerging from this analysis.

II. PROBLEM FORMULATION

This section introduces the agents, input representations and prediction scope that frame the statistical analysis carried out in the rest of this survey.

A. Agents in the Scene

In the context of motion forecasting, traffic participants are typically categorized into three distinct roles depending on their relationship to the prediction task. The **Ego Agent (EA)** is the autonomous vehicle carrying the onboard sensing and decision-making system. We refer to any traffic participant whose future motion is of interest as a **Target Agent (TA)**, the entity for which a trajectory must actually be forecast and around which the rest of the scene is structured. Notably, the EA itself may be designated as a TA, or it may act purely as an observer. Finally, **Surrounding Agent (SA)** comprise all other nearby traffic participants, whose past and present states provide the social and spatial context that conditions the Target Agent’s future behavior. The criteria used to select which nearby participants count as SA vary across the surveyed methods. Fig. 2 illustrates this relationship in a representative traffic scene.

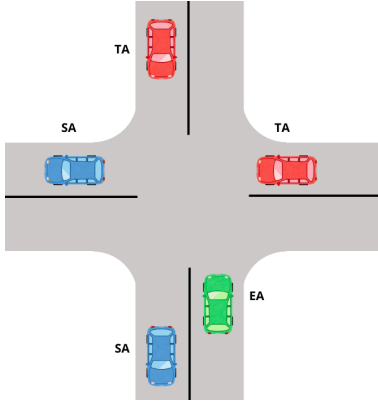


Fig. 2. Schematic illustration of the *Ego Agent* (EA), *Target Agents* (TA) and *Surrounding Agents* (SA) in a representative traffic scene.

B. Input Features

The way a scene is fed into a forecasting model varies considerably across the surveyed papers, both in terms of the overall input format and the specific features extracted.

At the format level, three broad strategies emerge, summarized in Table I. The dominant approach represents agents and map elements as *numeric*, *vectorized* inputs. Adopted by the vast majority of surveyed papers, this format utilizes structured arrays to encode positions, kinematic states and map geometry. A second, considerably smaller group of methods instead consumes *image* and *point-cloud* input, incorporating camera images and/or LiDAR point clouds directly into the forecasting

TABLE I
DISTRIBUTION OF THE REVIEWED STUDIES ACCORDING TO THE INPUT DATA FORMAT USED FOR TRAJECTORY PREDICTION.

Input format	# Papers	%
Numeric / vectorized	95	89.6%
Image / point-cloud	19	17.9%
Natural-language prompt	6	5.6%

pipeline. Finally, a rapidly expanding frontier in the field leverages Large Language Models (LLMs) by encoding the environment into *natural-language prompts*, translating numeric agent states and map context into text descriptions. These formats are not mutually exclusive, several papers combine vectorized inputs with auxiliary image or point-cloud features, so the statistical distributions presented in Table I exhibit overlapping categories.

Given this overall context, Fig. 3 summarizes the usage frequency of all input features identified across the surveyed papers.

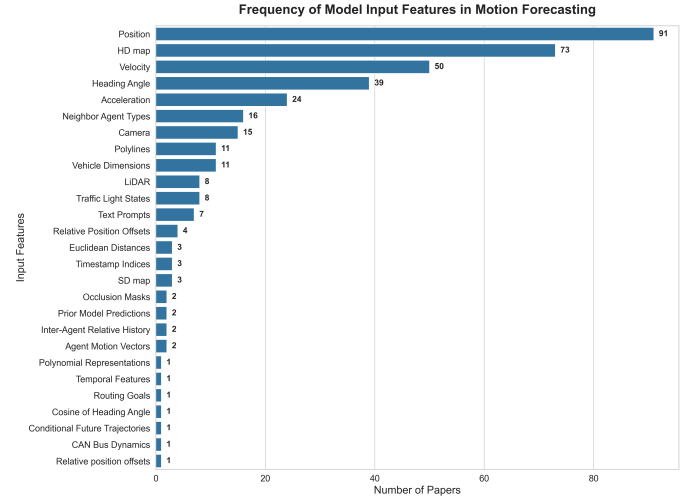


Fig. 3. Usage frequency of input features across the surveyed papers. Position, HD map information, velocity and heading angle are the four most widely adopted features.

As shown in Fig. 3, four input features clearly dominate: agent *position*, *HD map* information, *velocity* and *heading angle*. This distribution follows directly from what each feature represents and how costly it is to obtain. Position is the variable that most common evaluation metrics operate on directly, making it an almost unavoidable input regardless of the underlying architectural paradigm. Velocity and heading angle sit close behind because both are trivially derived from consecutive position observations via finite differences, so including them alongside a raw position history imposes essentially no additional data-collection cost. In the case of HD map information, rather than being derived, its high adoption reflects data availability, since it is natively bundled with the four dominant public benchmarks identified in Section III, making it accessible to any paper built on top of them without further engineering. Following these primary inputs, usage

frequencies decrease. Features like agent type or traffic light states are utilized in a moderate number of approaches. In contrast, there is a long tail of more specialized data that requires information beyond what these standard benchmark releases provide, such as SD maps and CAN-bus signals, which are only found in a handful of papers.

C. Single-Agent vs. Joint Multi-Agent Prediction

A further axis along which surveyed methods differ is the scope of the prediction itself.

- 1) **Single-agent** approaches forecast the future trajectory of one TA in isolation, as depicted conceptually in Fig. 4, treating SA only as conditioning context.
- 2) **Joint multi-agent** approaches instead forecast the future trajectories of the TAs, within a single model and a single forward pass, explicitly capturing the interactions and dependencies between agents' future behaviors.

As summarized in Table II, joint multi-agent prediction is considerably more common in the recent literature than single-agent prediction.

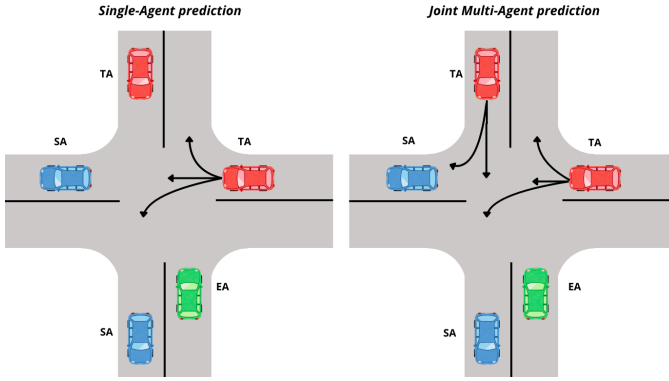


Fig. 4. Conceptual overview of motion prediction scope. Single-agent approaches predict only a chosen target. Joint multi-agent approaches simultaneously predict the future trajectories of multiple TAs.

TABLE II
DISTRIBUTION OF SURVEYED STUDIES ACCORDING TO THE PREDICTION TARGET.

Prediction Target	Number of Papers
Single-agent	28
Joint multi-agent	78

III. DATASETS

The formulation of the motion forecasting problem and the selection of target features are directly linked to the data collection platforms. To rigorously validate algorithmic performance and ensure cross-method comparability, the community relies on diverse real-world driving logs. A summary of the four most widely adopted public motion forecasting benchmarks identified in this survey is shown in Table III. The remainder of the surveyed papers rely on a long tail of less frequently used public benchmarks or on privately collected/simulated data, grouped under *Others* in the table; this

category includes, for instance, custom closed-loop simulation environments used to evaluate prediction-failure metrics rather than to benchmark forecasting accuracy directly.

- The **Argoverse** motion forecasting dataset [9] was released by Argo AI and provides large-scale, HD-map-annotated urban driving scenarios for 3D tracking and trajectory forecasting. Its successor, **Argoverse 2** [93], substantially expands the original release with 250,000 scenarios collected across six U.S. cities, each providing a local vector map together with 11 seconds of agent trajectory data sampled at 10 Hz, split into a 5-second observation window and a 6-second forecast horizon.
- The **Waymo Open Motion Dataset (WOMD)** [20] comprises more than 100,000 scenes, each 20 seconds long and sampled at 10 Hz, amounting to over 570 hours of driving data across more than 1,750 km of roadway, making it one of the largest public benchmarks in the field.
- The **nuScenes** [7] dataset contributes 1,000 driving scenes, each 20 seconds long and annotated with 3D bounding boxes at 2 Hz, collected across varied weather conditions and times of day in Boston and Singapore, using a full 360° sensor suite of six cameras, five radars and one LiDAR.
- The **INTERACTION** dataset [107] emphasizes densely interactive scenarios, recording vehicle trajectories at roundabouts, signalized and unsignalized intersections and merging/lane-change sites; across these scenario types it totals over 41,000 vehicle trajectories spanning approximately 990 minutes of driving behavior.

Beyond the four main benchmarks, a long tail of public and custom datasets have been used in a small number of studies. *Lyft Level 5* [27] appears in three papers [65], [97], [83]. Datasets such as *V2X-Seq* [105], *OSM-derived road network data* [4], *DRAMA* [55], *inD* [6], *V2V4Real* [98], *PEMS* [10], *CHBike* and *NYCTaxi* each appear in only one of the surveyed papers.

The characteristics of the datasets discussed above not only influence the evaluation benchmarks but also determine the temporal configurations adopted by the surveyed methods. As illustrated in Fig. 5, the most common configuration consists of an input sequence of 20 frames, adopted by 39 papers, followed by 10 and 50 frame windows. Shorter horizons and longer histories exceeding 30 frames are rare.

This concentration is largely explained by the characteristics of the dominant public benchmarks. For example, both *Argoverse 2* and the *Waymo Open Motion Dataset* provide trajectories sampled at 10 Hz with standardized observation windows, leading many authors to preserve the original temporal configuration rather than introducing alternative history lengths.

A similar trend can be observed for the prediction horizon, shown in Fig. 6. Forecasts of 30 frames represent the predominant choice, appearing in 34 surveyed papers, while prediction horizons of 60 and 80 frames are also frequently adopted. In contrast, intermediate or shorter horizons are only occasionally

TABLE III
OVERVIEW OF DATASETS UTILIZED ACROSS ANALYZED MODELS

Dataset	References	Count
Argoverse	[19], [32], [57], [100], [70], [82], [116], [74], [25], [86], [117], [4], [88], [119], [40], [67], [66], [96], [3], [104], [15], [17], [38], [11], [87], [14], [23], [115], [16], [91], [108], [79], [48], [112], [77], [42], [62], [94], [22], [51], [85], [18], [71], [103], [63], [5], [109], [45], [61], [47], [84], [118], [41], [81]	54
nuScenes	[106], [19], [57], [100], [65], [30], [26], [46], [69], [110], [73], [67], [64], [17], [14], [23], [8], [115], [16], [91], [108], [48], [54], [21], [97], [22], [28], [83], [45], [84], [39], [56], [58], [50], [44]	35
Waymo	[57], [70], [59], [65], [36], [66], [120], [33], [102], [76], [23], [91], [54], [95], [112], [77], [62], [31], [97], [43], [111], [113], [61], [101], [35], [34], [78], [41], [114], [81]	30
INTERACTION	[82], [65], [74], [96], [3], [35]	6
Others	[65], [97], [83], [72], [89], [13], [80], [99], [12], [1], [4], [58], [49], [75], [53], [52], [68], [92], [60], [24], [37]	21

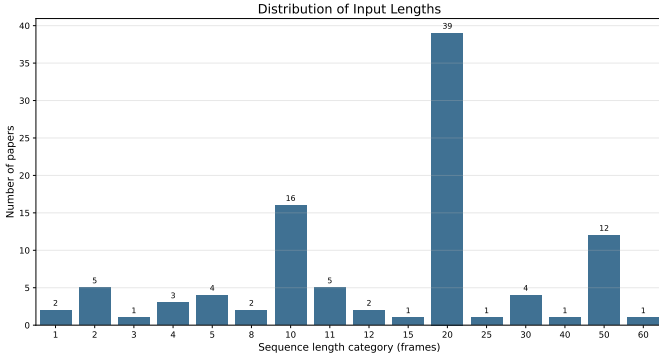


Fig. 5. Frequency of input sequence lengths adopted by the surveyed motion forecasting methods.

considered. This distribution again reflects the influence of widely used benchmarks, whose evaluation protocols define fixed forecasting intervals. For instance, datasets sampled at 10 Hz commonly evaluate predictions over 3 s (30 frames), 6 s (60 frames) or longer horizons, encouraging researchers to adopt these standardized settings. As a result, the surveyed literature exhibits a clear convergence towards benchmark-defined prediction horizons, improving comparability across different forecasting methods.

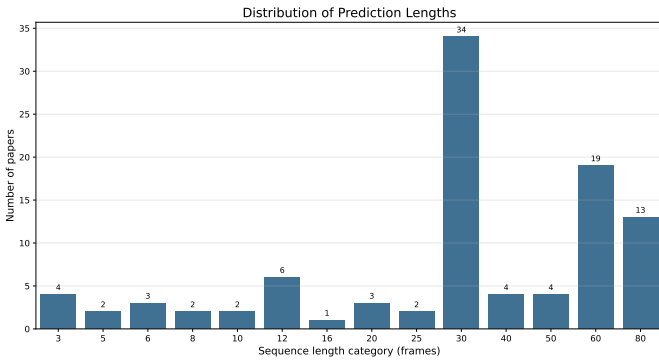


Fig. 6. Frequency of prediction horizons adopted by the surveyed motion forecasting methods.

IV. METRICS

To rigorously assess and compare the capabilities of various data-driven models, it is fundamental to employ consistent evaluation frameworks and standardized quantitative metrics, which allow different models to be compared with each other from diverse perspectives. In the domain of motion forecasting, modern methodologies primarily treat the problem as a multi-modal trajectory regression task to account for the inherent uncertainty and probabilistic nature of traffic scenes, as shown in Table II. The vast majority of works focus on capturing complex multi-agent interactions rather than isolated single-agent configurations.

As summarized in Fig. 7, multi-modal extensions of displacement errors, specifically Minimum Average Displacement Error (minADE_K) and Minimum Final Displacement Error (minFDE_K), form the absolute cornerstone of model validation in the surveyed papers. Furthermore, to evaluate how well models capture multi-modality without unfairly penalizing plausible alternative choices, the Miss Rate (MR) and the Best-of-K Minimum Final Displacement Error ($b\text{-minFDE}$) are widely adopted. This trend is highly correlated with their usage as primary ranking criteria in standardized benchmarks such as WOMD, Argoverse and nuScenes.

We formally define the four most prevalent metrics identified in Fig. 7. For the following formulations, let T denote the total future prediction horizon timesteps, $y_t \in \mathbb{R}^2$ represent the ground-truth coordinates of the target agent at time t and K represent the number of multi-modal trajectory hypotheses generated by the model. In simple terms, T rules how far into the future we are predicting, y_t denotes the exact path the vehicle actually took in the real world and K represents the different alternative futures the model proposes to account for uncertainty.

A. Minimum Average Displacement Error (minADE_K)

In multi-modal forecasting, a model outputs a set of K alternative future trajectories to cover different potential maneuvers. Standard deterministic metrics would unfairly penalize correct alternative guesses. To address this, minADE_K evaluates the model by selecting only the single closest trajectory out of the K predictions relative to the ground truth, averaging its Euclidean distance over all timesteps:

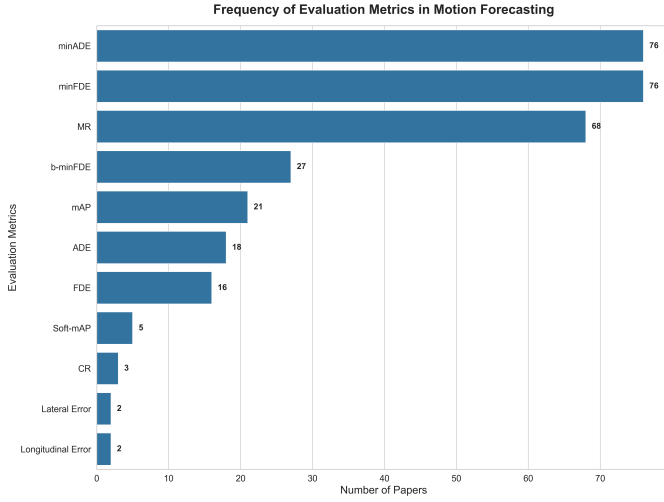


Fig. 7. Frequency of evaluation metrics across the surveyed motion forecasting literature.

$$\text{minADE}_K = \min_{k \in \{1, \dots, K\}} \frac{1}{T} \sum_{t=1}^T \|y_t - \hat{y}_t^k\|_2 \quad (1)$$

where $\hat{y}_t^k$ represents the predicted coordinates at time t for the k -th trajectory hypothesis.

B. Minimum Final Displacement Error (minFDE_K)

Following the same selective logic as minADE_K , the minFDE_K evaluates multi-modal performance strictly at the final milestone of the prediction horizon ($t = T$). It identifies the single best trajectory among the K proposals based solely on which one finishes closest to the true endpoint, ignoring any intermediate deviations:

$$\text{minFDE}_K = \min_{k \in \{1, \dots, K\}} \|y_T - \hat{y}_T^k\|_2 \quad (2)$$

This metric is critical for long-term downstream driving planning, as missing the final destination completely can lead to safety-critical failures.

C. Miss Rate (MR)

The Miss Rate complements displacement errors by calculating the proportion of test cases where the model fails to produce any acceptable prediction. A prediction is classified as a "miss" if the final displacement error of the best predicted trajectory exceeds a strict lateral or longitudinal threshold d_{thresh} . Formally, for a dataset of N scenes, it is defined as:

$$\text{MR} = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left(\min_k \|y_{T,i} - \hat{y}_{T,i}^k\|_2 > d_{\text{thresh}} \right) \quad (3)$$

where $\mathbb{I}(\cdot)$ is the indicator function that outputs 1 if the condition is true and 0 otherwise.

D. Best-of- K Minimum Final Displacement Error (b-minFDE)

The b -minFDE is a variation utilized in highly complex multi-modal evaluation pipelines. It measures the minimum final displacement error over the best subset of the predicted configurations, ensuring that the model's highest-confidence trajectory profiles are strictly conformed to the topological constraints of the road. It can be formulated as:

$$b\text{-minFDE} = \min_{k \in \mathcal{B}_K} \|y_T - \hat{y}_T^k\|_2 \quad (4)$$

where $\mathcal{B}_K$ represents the optimized set of the top- K scoring multi-modal trajectories ranked by the model's classification or probability head.

V. MODEL ARCHITECTURES

While the choice of evaluation metrics provides a standardized protocol to quantify performance, the architectural mechanisms designed to optimize these metrics vary substantially across the literature. In order to address this variety of architectures, Fig. 8 presents the architectural taxonomy used to organize the models surveyed in this section. We group the surveyed papers into four families, ordered from the most classical to the most recent: physics-based and rule-based models, reinforcement learning, deep learning and large language models. The deep learning family, which dominates the recent literature, is further broken down into eight architecture classes, discussed individually below.

A. Physics-based & Rule-based Models

The earliest and most interpretable family of approaches grounds trajectory prediction directly in vehicle kinematics or explicit driving rules, rather than learning the underlying motion dynamics purely from data. *Deep Stochastic Kinematic Models* [113] exemplifies the purely physics-based end of this family: instead of letting a neural network implicitly learn how positional uncertainty grows over time, it derives this growth analytically from a bicycle kinematic model, injecting physical prior knowledge directly into the probabilistic forecast. A related line of work represents trajectories and road geometry through closed-form mathematical functions rather than raw point sequences, for instance using Bernstein polynomials to improve out-of-distribution generalization across unseen cities and sensor configurations [103], or adversarially perturbing scenes through simple, interpretable physical transformations to expose the limited generalization of learning-based predictors [4].

A second, common branch combines explicit rules or driving-style priors with a learned component, rather than relying on physics alone. *Multi-Predictor Fusion* [83] makes this hybridization explicit, fusing the output of a learned predictor (Trajectron++) with a rule-based, horizon-aware planner to obtain trajectories that are simultaneously data-driven and dynamically feasible. In a similar spirit, *SAH-Drive* [21] couples a rule-based planning core with a deep learning module through a biologically inspired, dual-timescale decision mechanism, explicitly aiming to combine the efficiency of

■ Model Family ■ Architecture class ■ Learning strategy

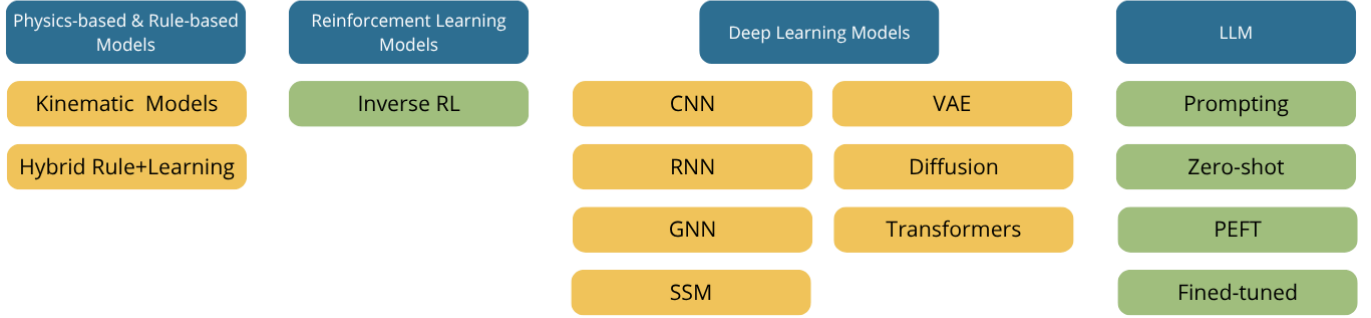


Fig. 8. Hierarchical taxonomy of motion forecasting methodologies analyzed in this survey. The literature is structured into distinct algorithmic families based on their core architectural designs, spanning traditional physics-based baselines, sequential networks, graph-based architectures, attention mechanisms and recent generative paradigms such as diffusion models and large language models.

fixed rules with the flexibility of learned behavior. Driving-style priors are leveraged in [91], which classifies drivers into conservative, normal and aggressive styles and adapts its kinematic basis functions accordingly, while [13] combines a GNN-based intent recognition module with an Unscented Kalman Filter (UKF) to constrain predicted trajectories to be kinematically feasible and resolves right-of-way conflicts at unsignalized intersections.

B. Reinforcement Learning

A small but conceptually distinct line of work frames trajectory prediction as an inverse planning problem, rather than as direct sequence regression. *Foresight in Motion* [67] exemplifies this perspective: rather than mapping observed states directly to future coordinates, it first infers a reward function that explains an agent’s observed behavior through Maximum-Entropy Inverse Reinforcement Learning and only then generates a trajectory consistent with that inferred intent. This reward-driven, planning-oriented formulation stands apart from the predominantly feed-forward, regression-style methods that dominate the rest of the surveyed literature, offering a complementary, more behaviorally grounded route toward motion forecasting that remains comparatively underexplored in recent work.

C. Deep Learning Models

Deep learning methods dominate the surveyed literature by a wide margin, spanning a diverse set of architectural paradigms. this section is organized into eight architecture classes, following the taxonomy summarized in Fig. 8.

Convolutional Neural Networks (CNNs). Convolutional backbones remain a competitive alternative to graph-based and attention-based designs, particularly when the input is represented as a rasterized bird’s-eye-view (BEV) image rather than a vectorized scene. [80] shows that a carefully designed two-stage CNN can model agent interactions as effectively as more complex graph-based approaches, while operating substantially faster. CNNs are also the natural backbone when

forecasting is coupled directly to a perception stack. *Future-Det* [69] fuses detection and forecasting into a single pipeline that operates directly on raw LiDAR sweeps and [46] similarly performs class-agnostic motion prediction directly over BEV occupancy grids in a weakly supervised setting.

Recurrent Neural Networks (RNNs). Rather than serving as a standalone forecasting backbone, recurrent units persist in the recent literature as a lightweight sequential component integrated into architectures that are based on graphs or attention mechanisms. *FJMP* [74] relies on a GRU both to encode each agent’s trajectory history and to update interacting agents’ features as information propagates through its learned interaction graph, while *PAGA* [18] uses an LSTM to encode sequential lane-connectivity features feeding into its graph attention module. LSTMs also appear at multiple stages of more elaborate multi-agent systems. For instance, [41] uses one LSTM to generate intermediate natural-language hypotheses about agent intent and another, with agent and token level attention, to decode the final trajectory conditioned on those hypotheses. RNN-based predictors also remain a relevant point of comparison for robustness studies. In [110] specifically probes the adversarial robustness of LSTM-based FQA and GRU-based GRIP++. This pattern suggests that, while pure RNN-based forecasting has largely been replaced by attention-based and graph-based encoders, recurrent units continue to play a supporting role wherever a compact, order-aware representation of a sequence, whether a trajectory history or a chain of connected lanes is required.

Graph Neural Networks (GNNs). GNNs represent the driving scene explicitly as an interaction graph between agents and map elements and remain the most natural alternative to Transformer-based encoding for modeling multi-agent dependencies. *PAGA* [18] extends graph attention specifically to capture long-range relationships between non-adjacent lanes in HD maps. Building on this map-centric use of graphs, *FJMP* [74] instead applies graph-based reasoning to the agents themselves, modeling their future interactions as a learned

directed acyclic graph that captures influencer-reactor relationships between them. *HDTG* [35] unifies both directions by treating every agent, lane and traffic signal as a heterogeneous node connected through typed edges within a single graph. Most recently, *BeTop* [54] refines interaction modeling further, formulating joint future interactions as a topological “braid set” over the interaction graph rather than a fixed set of pairwise edges.

Beyond forecasting accuracy itself, GNN encoders also serve as reusable building blocks for problems adjacent to core prediction. [99] tackles trajectory imputation and prediction jointly, coupling a multi-space GNN, which combines static, dynamically learned and edge-conditioned graph convolutions to handle incomplete observations. [47] instead uses a GNN-based backbone, DenseTNT with a VectorNet encoder, as the target of a large-scale pretraining pipeline. The model is pretrained with a masked-autoencoding objective on hundreds of thousands of synthetic maps before fine-tuning on real data. Finally, [12] extends single-view GNN encoders such as HiVT or LaneGCN to a cooperative, multi-view setting, fusing vehicle and infrastructure graph embeddings through cross-graph attention and calibrating the resulting multimodal predictions with a post-hoc conformal prediction module that provides statistically rigorous uncertainty intervals.

State Space Models. A very recent and still narrow line of work replaces attention with selective state space models (SSMs) to address the quadratic complexity of Transformer-based encoders. *Trajectory-Mamba* [32] is, to our knowledge, the only representative of this class in the surveyed literature, combining a selective SSM with lightweight attention to reduce the encoding complexity from quadratic to linear in the number of agents and time steps, while retaining competitive forecasting accuracy.

Variational Autoencoders (VAEs). VAE-based methods address the inherently multimodal nature of future motion through an explicit latent variable model, rather than relying solely on multiple regression heads or anchors. [16] introduces a hierarchical latent structure specifically designed to mitigate “mode blur”, where competing plausible futures collapse into an unrealistic averaged trajectory. [97] instead introduces latent variables at every time step through a timewise VAE formulation, coupled with a dual attention mechanism, to model the evolving uncertainty of human decisions under strict real-time constraints. A related, adversarially regularized variant is adopted in [38], where an adversarial autoencoder built on top of a LaneGCN backbone structures its latent space into semantically meaningful longitudinal, lateral and residual components, explicitly separating discrete driving intentions from the continuous trajectory itself to improve robustness when labeled data is limited.

Diffusion and Flow-based Generative Models. Diffusion and flow-matching formulations have emerged as a competitive alternative for multimodal trajectory generation, approaching prediction as an iterative denoising or transport process rather than a single deterministic mapping. *Motion-Diffuser* [36] performs denoising in a PCA-compressed la-

tent trajectory space to enable fast sampling and tractable exact likelihood computation, while [87] further optimizes the diffusion process itself to make multi-agent, controllable generation practical under real-time constraints. Beyond raw forecasting accuracy, controllable diffusion models are also used generatively to synthesize rare, safety-critical scenarios for long-tail training [66]. Most recently, *TrajFlow* [101] replaces iterative denoising with flow matching, mapping noise to plausible trajectories through a learned ODE and predicting multiple future modes in a single sampling pass. Also, *TransDiffuser* [37] combines a frozen Transfuser perception backbone with a DDPM-style denoising decoder, explicitly decorrelating the latent representations feeding the diffusion process to encourage more diverse trajectory generation.

Transformers. Transformer-based architectures dominate the recent literature, to the point of constituting the default backbone for motion forecasting. Most Transformer-based designs in our corpus share a common architectural backbone, a self-attention encoder over agent and map tokens coupled with a cross-attention decoder that queries this encoded scene to produce future trajectories, while differing in how this backbone structures its attention computation. Three influential 2022 designs illustrate distinct ways of organizing this computation. *Scene Transformer* [62] encodes the entire scene in a shared, scene-centric coordinate frame to produce scene-consistent joint predictions. *HiVT* [119] instead adopts a hierarchical, agent-centric formulation for lightweight yet accurate multi-agent inference. *MTR* [77] combines global intention localization with local motion refinement through a limited, learnable set of intention queries. The following year brought two further refinements along different axes. *Wayformer* [61] shows that a simple, modality-agnostic self-attention encoder can match more specialized designs, establishing an efficient generic recipe that much of the subsequent literature adopts as a starting point, while *QCNet* [117] introduces a query-centric paradigm that avoids costly re-encoding of the scene as the observation window slides forward in time. This family of centric formulations was later extended by *SIMPL* [109], which adopts an instance-centric representation, establishing a local reference frame for every agent and lane segment individually before fusing them through a symmetric fusion Transformer. *MTR*’s intention-query design was itself extended two years later by *MTR++* [78], which adapts the same learnable intention points to symmetric, joint multi-agent scene modeling rather than single-agent prediction.

A distinct and rapidly growing branch within this family reformulates forecasting itself as autoregressive GPT-style sequence generation over discretized motion tokens. *MotionLM* [76] and *SMART* [95] both adapt the next-token prediction paradigm of large language models to continuous motion, while *DriveGPT* [31] and *DONUT* [40] push this formulation further by unifying past and future trajectory modeling within a single decoder-only architecture. This autoregressive formulation has also been extended toward end-to-end planning. *ARTEMIS* [24] generates a short planning trajectory through eight autoregressive steps, routing each step

through a mixture of experts and refining the result with cross-attention over a Transfuser-based camera-LiDAR perception backbone. A parallel line of work couples this Transformer-based decoding with a joint detection and forecasting formulation. *IMPACT* [81] combines a shared scene encoder with an MTR style cross-attention decoder to jointly refine an agent’s detected past motion and forecast its future, while *MASAR* [44] applies a similar detection and forecasting philosophy directly to multi-view camera images, using factorized temporal, modal and social attention to jointly refine past detections and predict multiple future modes. Given the huge volume and diversity of Transformer-based work in our corpus, spanning agent-centric, scene-centric, query-centric, instance-centric and autoregressive designs alike, this family alone accounts for the majority of all architectural innovation identified in this survey.

D. Large Language Models (LLMs)

The most recent architectural shift in the surveyed literature repurposes pretrained large language models as the forecasting or reasoning backbone, rather than training a specific network from scratch. These approaches differ not only in how the language model is adapted, through prompting or fine-tuning, but more fundamentally in the modality through which it actually receives the scene.

Most approaches convert the scene into a natural-language prompt and reason over text. *GPT-Driver* [56] reframes motion planning itself as a language modeling problem, tokenizing observations into natural-language descriptions and querying GPT-3.5 directly, without modifying its underlying weights. [39] similarly integrates pretrained language encoders directly into the trajectory prediction pipeline, training the full system end-to-end on textual scene descriptions rather than vectorized or image-based input. *LC-LLM* [68] instead fine-tunes Llama-2-13b-chat on textual scene descriptions using LoRA [29], a parameter-efficient adaptation technique that injects small trainable low-rank matrices while keeping the pretrained weights frozen, paired with explicit Chain-of-Thought reasoning [90].

A second, comparatively rare group instead feeds the language model a visual representation of the scene rather than a text description. [114] relies on systematic prompt engineering alone, introducing a rasterized Transportation Context Map to help an untrained GPT-4V reason directly over a bird’s-eye-view image of the traffic scene.

A third group moves away entirely from both textual and visual information, instead reuse the pretrained weights or training techniques of a language model to operate on purely numerical, vectorized scene representations. [53] feeds spatio-temporal traffic embeddings, not natural-language tokens, into a partially frozen GPT-2/LLaMA-2 backbone, repurposing its sequence modeling capacity for numerical forecasting. *Forecast-PEFT* [84] goes further, borrowing only the parameter-efficient fine-tuning (PEFT) paradigm popularized in the LLM literature and applying it to a Transformer autoencoder that processes vectorized agent trajectories and lane

segments, without involving a language model at inference at all.

Bridging these groups, *CoT-Drive* [50] adopts a teacher-student distillation strategy in which GPT-4 Turbo, prompted with textual scene descriptions, generates detailed Chain-of-Thought risk annotations that are distilled into a much smaller, deployable student model combining a lightweight language-instructed encoder with a Transformer-based spatial encoder. Taken together, these approaches suggest that LLMs are being explored less as a full replacement for specialized forecasting architectures and more as a source of contextual reasoning, explainability and common-sense understanding, whether accessed through text, vision, or simply an inherited architectural backbone, that complements them.

VI. DISCUSSION AND OPEN GAPS

The statistical and architectural picture that has emerged throughout this survey shows a field that has matured rapidly, converging on shared benchmarks, shared evaluation protocols and a common architectural vocabulary centered on transformers. This convergence, however, has left several aspects of the problem relatively unanalysed. This section discusses such gaps, each grounded in patterns already surfaced earlier in this survey and illustrated through specific works in our corpus that directly address them.

A. Evaluation Metrics and Real-World Reliability

The metrics reviewed in Section IV strongly support reward point-estimate distance accuracy, rewarding the single closest of several predicted modes to the ground truth. None of the metrics adopted across the surveyed papers report a calibrated measure of predictive confidence, such as a likelihood or a calibration error, alongside this distance based accuracy. *Regret* [60] makes the resulting disconnect explicit, showing that a predictor can score well on standard distance metrics while nonetheless causing a subsequent planning failure, because the metric evaluates each prediction in isolation rather than in the context of the decision it ultimately informs. Similarly, citewiederer2023joint highlights a related blind spot, treating out-of-distribution detection and uncertainty estimation as a dedicated additional module layered on top of an otherwise standard predictor, rather than as a property most predictors are expected to have by default. Taken together, these works suggest that the field’s benchmarking culture currently rewards proximity to a single ground-truth future, a proxy that does not necessarily correlate with the reliability that safety-critical deployment actually requires.

B. Perception Robustness

As shown in Fig. 1, the large majority of the surveyed literature operates on structured, vectorized scene representations rather than on raw sensor data, implicitly assuming that upstream detection and tracking have already resolved each agent’s past trajectory with near-perfect accuracy. A small but distinct cluster of works probes what happens once this assumption is relaxed. [106], [110] and [8] each craft realistic

perturbations, whether adversarial or naturalistic, to already-tracked trajectories and observe substantial degradation in otherwise strong predictors, while [4] shows a complementary form of brittleness under structural, map-level perturbation rather than agent-level noise. The fact that this kind of stress-testing remains confined to a small, self-identified sub-community, rather than forming a standard part of how any new predictor is evaluated, suggests that the field’s reported accuracy figures may overstate how well these methods would hold up against the imperfect perception a deployed vehicle actually has to work with.

C. Underexploited Contextual and Cooperative Signals

Section II-B shows that a handful of features, position, HD map information, velocity and heading angle, account for the large majority of surveyed input designs, while richer contextual signals form a long tail of comparatively rare inputs. This narrowness is notable given that considerably more context is, in principle, already available. Cooperative-perception datasets that share information across vehicles and infrastructure, such as *V2X-Seq* [105] and *V2V4Real* [98], are each adopted by only a single work in our corpus, [12] and [89] respectively and a third such dataset tracked in our review is not adopted by any surveyed paper at all. Even comparatively simple, already-standardized infrastructure information, such as traffic light state, appears in well under a tenth of the surveyed works. This suggests that closing the gap between the contextual richness a real, connected vehicle could in principle access and the narrow input recipe most current predictors are designed around may be as consequential for real-world performance as further architectural innovation on top of that same narrow recipe.

D. Generalization

The dataset landscape described in Section III is dominated by a small number of large public benchmarks, each tied to a limited set of cities and road layouts. [4] probes what happens outside this comfort zone, showing that several established predictors, near-perfect when evaluated on in-distribution road geometries, degrade sharply when tested on simple, interpretable perturbations of the road topology that never appear during training. This finding is difficult to reconcile with the field’s current reporting conventions, which, almost without exception, evaluate a model using validation data drawn from the same small number of cities and sensor configurations that were used for training. Whether the geographic and structural diversity already present across the standard benchmarks is sufficient to expose this kind of weakness, or whether dedicated cross-topology and cross-city evaluation protocols are needed to surface it reliably, remains an open methodological question.

E. Interpretability

Section V shows that Transformer-based designs, in their many encoder and decoder variants, account for the majority of the architectural innovation identified in this survey. This

dominant design family, like the graphs and convolutions architectures that preceded it, offers little insight into why a particular future was predicted, beyond the indirect, partial evidence that attention weights can provide. [28] is a rare exception that treats interpretability as a core design objective, deriving explanations from an interpretable Hidden Markov Model that explicitly identifies each agent’s contribution to the prediction, rather than relying on visualizations generated after the prediction process. The emerging use of large language models discussed in Section V offers a second, largely orthogonal route toward this same goal, valued less for matching the raw accuracy of specialized Transformers and more for the natural language justifications they can attach to a prediction. That both of these directions remain comparatively rare suggests that interpretability is still treated as a specialized pursuit rather than a mainstream design requirement, even as the field’s dominant architectures grow increasingly difficult to inspect.

Across these five directions, a common pattern emerges. This survey has documented real progress, from improved accuracy on standard benchmarks to increasingly capable Transformer-based architectures and now the first steps toward using large language models. Yet most of this progress has happened within a fixed set of assumptions. The field still relies on a handful of benchmarks and a handful of distance-based metrics. One architectural family increasingly dominates the rest and most models are trained on a narrow set of input features. These same assumptions made the analysis in this survey possible, but they are also, as the gaps discussed above show, the boundary of what we currently understand well. Closing these gaps will likely take more than simply scaling up current models. It will instead require rethinking the benchmarks, the metrics, the architectures, the perception pipeline and the input features themselves, treating each as an open design choice rather than a fixed starting point and subject each of them to the same level of analysis that had previously been reserved for the models created on the basis of them.

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REFERENCES

- [1] M. E. Akbiyik, N. Savov, D. P. Paudel, N. Popovic, C. Vater, O. Hilliges, L. Van Gool, and X. Wang, “Leveraging driver field-of-view for multimodal ego-trajectory prediction,” in *ICLR*. OpenReview, 2025.
- [2] M. Antunes, L. M. Bergasa, S. Montiel-Marín, R. Barea, F. Sánchez-García, and A. Llamazares, “Fast and efficient transformer-based method for birds eye view instance prediction,” in *ITSC*. IEEE, 2024, pp. 1269–1274.
- [3] G. Aydemir, A. K. Akan, and F. Güney, “Adapt: Efficient multi-agent trajectory prediction with adaptation,” in *ICCV*. IEEE, 2023, pp. 8261–8271.

- [4] M. Bahari, S. Saadatnejad, A. Rahimi, M. Shaverdikondori, A. H. Shahidzadeh, S.-M. Moosavi-Dezfooli, and A. Alahi, "Vehicle trajectory prediction works, but not everywhere," in *CVPR*. IEEE, 2022, pp. 17 102–17 112.
- [5] P. Bhattacharyya, C. Huang, and K. Czarnecki, "SSL-Lanes: Self-supervised learning for motion forecasting in autonomous driving," in *CoRL*, vol. 205. PMLR, 2022, pp. 1793–1805.
- [6] J. Bock, R. Krajewski, T. Moers, L. Vater, S. Runde, C. Homeier, and L. Eckstein, "The ind dataset: A drone dataset of naturalistic road user trajectories at german intersections," in *IV*. IEEE, 2020, pp. 1929–1934.
- [7] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuscenes: A multimodal dataset for autonomous driving," in *CVPR*. IEEE, 2020, pp. 11 618–11 628.
- [8] Y. Cao, C. Xiao, A. Anandkumar, D. Xu, and M. Pavone, "AdvDO: Realistic adversarial attacks for trajectory prediction," in *ECCV*. Springer, 2022, pp. 36–52.
- [9] M.-F. Chang, J. Lambert, P. Sangkloy, J. Singh, S. Bak, A. Hartnett, D. Wang, P. Carr, S. Lucey, D. Ramanan, and J. Hays, "Argoverse: 3d tracking and forecasting with rich maps," in *CVPR*. IEEE, 2019, pp. 8740–8749.
- [10] C. Chen, K. Petty, A. Skabardonis, P. Varaiya, and Z. Jia, "Freeway performance measurement system: Mining loop detector data," *Transportation Research Record*, vol. 1748, no. 1, pp. 96–102, 2001.
- [11] H. Chen, J. Wang, K. Shao, F. Liu, J. Hao, C. Guan, G. Chen, and P.-A. Heng, "Traj-mae: Masked autoencoders for trajectory prediction," in *ICCV*. IEEE, 2023, pp. 8317–8328.
- [12] X. Chen, R. Bhadani, and L. Head, "Conformal trajectory prediction with multi-view data integration in cooperative driving," *SSRN Electronic Journal*, 2025.
- [13] Z. Chen, P. Chen, S. Yang, L. Li, J. Wu, R. Sun, Z. Wang, and G. Yu, "Multi-modal vehicle trajectory prediction via hierarchical attention and raster-vector maps encoding in unstructured road environments," *EEAI*, vol. 167, p. 113819, 2026.
- [14] Z. Chen, M. Ye, S. Xu, T. Cao, and Q. Chen, "Ppad: Iterative interactions of prediction and planning for end-to-end autonomous driving," in *ECCV*. Springer, 2024, pp. 239–256.
- [15] J. Cheng, X. Mei, and M. Liu, "Forecast-mae: Self-supervised pre-training for motion forecasting with masked autoencoders," in *ICCV*. IEEE, 2023, pp. 8645–8655.
- [16] D. Choi and K. Min, "Hierarchical latent structure for multi-modal vehicle trajectory forecasting," in *ECCV*. Springer, 2022, pp. 129–145.
- [17] S. Choi, J. Kim, J. Yun, and J. W. Choi, "R-pred: Two-stage motion prediction via tube-query attention-based trajectory refinement," in *ICCV*. IEEE, 2023, pp. 8491–8501.
- [18] F. Da and Y. Zhang, "Path-aware graph attention for HD maps in motion prediction," in *ICRA*. IEEE, 2022, pp. 6430–6436.
- [19] Z. Dong, R. Ding, W. Li, P. Zhang, G. Tang, and J. Guo, "Leveraging sd map to augment hd map-based trajectory prediction," in *CVPR*. IEEE, 2025, pp. 17 219–17 228.
- [20] S. Ettinger, S. Cheng, B. Caine, C. Liu, H. Zhao, S. Pradhan, Y. Chai, B. Sapp, C. R. Qi, Y. Zhou, Z. Yang, A. Chouard, P. Sun, J. Ngiam, V. Vasudevan, A. McCauley, J. Shlens, and D. Anguelov, "Large scale interactive motion forecasting for autonomous driving: The waymo open motion dataset," in *ICCV*. IEEE, 2021, pp. 9710–9719.
- [21] Y. Fan, Z. Cui, Z. Li, Y. Ren, and H. Yu, "Sah-drive: A scenario-aware hybrid planner for closed-loop vehicle trajectory generation," in *ICML*. OpenReview, 2025.
- [22] C. Feng, H. Zhou, H. Lin, Z. Zhang, Z. Xu, C. Zhang, B. Zhou, and S. Shen, "MacFormer: Map-agent coupled transformer for real-time and robust trajectory prediction," *RA-L*, vol. 8, no. 10, pp. 6795–6802, 2023.
- [23] L. Feng, M. Bahari, K. M. B. Amor, E. Zablocki, M. Cord, and A. Alahi, "Unitraj: A unified framework for scalable vehicle trajectory prediction," in *ECCV*. Springer, 2024, pp. 106–123.
- [24] R. Feng, N. Xi, D. Chu, R. Wang, Z. Deng, A. Wang, L. Lu, J. Wang, and Y. Huang, "Artemis: Autoregressive end-to-end trajectory planning with mixture of experts for autonomous driving," *RA-L*, vol. 11, no. 1, pp. 226–233, 2026.
- [25] C. Gomez-Huélamo, M. V. Conde, R. Barea, and L. M. Bergasa, "Improving multi-agent motion prediction with heuristic goals and motion refinement," in *CVPRW*. IEEE, 2023, pp. 5323–5332.
- [26] J. Gu, C. Hu, T. Zhang, X. Chen, Y. Wang, Y. Wang, and H. Zhao, "Vip3d: End-to-end visual trajectory prediction via 3d agent queries," in *CVPR*. IEEE, 2023, pp. 5496–5506.
- [27] J. Houston, G. Zuidhof, L. Bergamini, Y. Ye, A. Jain, S. Omari, V. Iglovikov, and P. Ondruska, "One thousand and one hours: Self-driving motion prediction dataset," in *CoRL*. PMLR, 2020.
- [28] K.-C. Hsu, K. Leung, Y. Chen, J. F. Fisac, and M. Pavone, "Interpretable trajectory prediction for autonomous vehicles via counterfactual responsibility," in *IROS*. IEEE, 2023, pp. 5918–5925.
- [29] E. J. Hu, yelong shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, "Lora: Low-rank adaptation of large language models," in *ICLR*. OpenReview, 2022.
- [30] Y. Hu, J. Yang, L. Chen, K. Li, C. Sima, X. Zhu, S. Chai, S. Du, T. Lin, W. Wang, L. Lu, X. Jia, Q. Liu, J. Dai, Y. Qiao, and H. Li, "Planning-oriented autonomous driving," in *CVPR*. IEEE, 2023, pp. 17 853–17 862.
- [31] X. Huang, E. M. Wolff, P. Vernaza, T. Phan-Minh, H. Chen, D. S. Hayden, M. Edmonds, B. Pierce, X. Chen, P. E. Jacob, X. Chen, C. Tairbekov, P. Agarwal, T. Gao, Y. Chai, and S. Srinivasa, "DriveGPT: Scaling autoregressive behavior models for driving," in *PMLR*, vol. 267. PMLR, 2025, pp. 25 908–25 921.
- [32] Y. Huang, Y. Cheng, and K. Wang, "Trajectory mamba: Efficient attention-mamba forecasting model based on selective ssm," in *CVPR*. IEEE, 2025, pp. 12 058–12 067.
- [33] Z. Huang, H. Liu, and C. Lv, "Gameformer: Game-theoretic modeling and learning of transformer-based interactive prediction and planning for autonomous driving," in *ICCV*. IEEE, 2023, pp. 3880–3890.
- [34] X. Jia, S. Shi, Z. Chen, L. Jiang, W. Liao, T. He, and J. Yan, "AMP: Autoregressive motion prediction revisited with next token prediction for autonomous driving," *arXiv preprint arXiv:2403.13331*, 2024.
- [35] X. Jia, P. Wu, L. Chen, Y. Liu, H. Li, and J. Yan, "Hdgt: Heterogeneous driving graph transformer for multi-agent trajectory prediction via scene encoding," *TPAMI*, vol. 45, no. 11, pp. 13 860–13 875, 2023.
- [36] C. Jiang, A. Cornman, C. Park, B. Sapp, Y. Zhou, and D. Anguelov, "Motiondiffuser: Controllable multi-agent motion prediction using diffusion," in *CVPR*. IEEE, 2023, pp. 9644–9653.
- [37] X. Jiang, Y. Ma, P. Li, L. Xu, X. Wen, K. Zhan, Z. Xia, P. Jia, X. Lang, and S. Sun, "Transdiffuser: Diverse trajectory generation with decorrelated multi-modal representation for end-to-end autonomous driving," *arXiv preprint arXiv:2505.09315*, 2026.
- [38] R. Jiao, X. Liu, T. Sato, Q. A. Chen, and Q. Zhu, "Semi-supervised semantics-guided adversarial training for robust trajectory prediction," in *ICCV*. IEEE, 2023, pp. 8207–8217.
- [39] A. Keysan, A. Look, E. Kosman, G. Gursun, J. Wagner, Y. Yao, and B. Rakitsch, "Can you text what is happening? Integrating pre-trained language encoders into trajectory prediction models for autonomous driving," *CoRR*, vol. abs/2309.05282, 2023.
- [40] M. Knoche, D. de Geus, and B. Leibe, "Donut: A decoder-only model for trajectory prediction," in *ICCV*. IEEE, 2025, pp. 28 903–28 912.
- [41] Y.-L. Kuo, X. Huang, A. Barbu, S. G. McGill, B. Katz, J. J. Leonard, and G. Rosman, "Trajectory prediction with linguistic representations," in *ICRA*. IEEE, 2022, pp. 2868–2875.
- [42] Z. Lan, Y. Jiang, Y. Mu, C. Chen, and S. E. Li, "Sept: Towards efficient scene representation learning for motion prediction," in *ICLR*. OpenReview, 2024.
- [43] B. Lange, J. Li, and M. J. Kochenderfer, "Scene informer: Anchor-based occlusion inference and trajectory prediction in partially observable environments," in *ICRA*. IEEE, 2024, pp. 14 138–14 145.
- [44] M. A. B. Lehocine, J. Schmidt, F. Moosmann, D. Gupta, and F. Flohr, "Masar: Motion-appearance synergy refinement for joint detection and trajectory forecasting," in *ICRA*. IEEE, 2026.
- [45] R. Li, C. Li, Y. Li, H. Li, Y. Chen, Y. Yuan, and G. Wang, "Itpnet: Towards instantaneous trajectory prediction for autonomous driving," in *KDD*. ACM, 2024, pp. 1643–1654.
- [46] R. Li, H. Shi, Z. Fu, Z. Wang, and G. Lin, "Weakly supervised class-agnostic motion prediction for autonomous driving," in *CVPR*. IEEE, 2023, pp. 17 599–17 608.
- [47] Y. Li, S. Z. Zhao, C. Xu, C. Tang, C. Li, M. Ding, M. Tomizuka, and W. Zhan, "Pre-training on synthetic driving data for trajectory prediction," in *IROS*. IEEE, 2024, pp. 5910–5917.
- [48] Y. Li, C. Li, R. Lv, R. Li, Y. Yuan, and G. Wang, "Lakd: Length-agnostic knowledge distillation for trajectory prediction with any length observations," in *NeurIPS*, vol. 37. Curran Associates, Inc., 2024, pp. 28 720–28 744.

- [49] W. Liang, W. Cao, C. Wang, Z. Zhang, W. Zhang, and W. Zhao, "Game-theoretic motion planning method for dynamic interactive vehicles based on intent-awareness and multimodal trajectory prediction," *Advanced Engineering Informatics*, vol. 71, p. 104383, 2026.
- [50] H. Liao, H. Kong, B. Wang, C. Wang, K. Y. Wang, Z. He, C. Xu, and Z. Li, "CoT-Drive: Efficient motion forecasting for autonomous driving with llms and chain-of-thought prompting," *TAI*, vol. 7, no. 2, pp. 625–641, 2026.
- [51] H. Liao, S. Liu, Y. Li, Z. Li, C. Wang, Y. Li, S. E. Li, and C. Xu, "Human observation-inspired trajectory prediction for autonomous driving in mixed-autonomy traffic environments," in *ICRA*. IEEE, 2024, pp. 14 212–14 219.
- [52] —, "Human observation-inspired trajectory prediction for autonomous driving in mixed-autonomy traffic environments," in *ICRA*. IEEE, 2024, pp. 14 212–14 219.
- [53] C. Liu, S. Yang, Q. Xu, Z. Li, C. Long, Z. Li, and R. Zhao, "Spatial-temporal large language model for traffic prediction," in *MDM*. IEEE, 2024, pp. 31–40.
- [54] H. Liu, L. Chen, Y. Qiao, C. Lv, and H. Li, "Reasoning multi-agent behavioral topology for interactive autonomous driving," in *NeurIPS*, vol. 37. Curran Associates, Inc., 2024, pp. 92 605–92 637.
- [55] S. Malla, C. Choi, I. Dwivedi, J. H. Choi, and J. Li, "Drama: Joint risk localization and captioning in driving," in *WACV*. IEEE, 2023, pp. 1043–1052.
- [56] J. Mao, Y. Qian, J. Ye, H. Zhao, and Y. Wang, "GPT-DRIVER: Learning to drive with GPT," *CoRR*, vol. abs/2310.01415, 2023.
- [57] K. Messaoud, M. Cord, and A. Alahi, "Towards generalizable trajectory prediction using dual-level representation learning and adaptive prompting," in *CVPR*. IEEE, 2025, pp. 27 564–27 574.
- [58] S. Moon, H. Woo, H. Park, H. Jung, R. Mahjourian, H.-g. Chi, H. Lim, S. Kim, and J. Kim, "Visiontrap: Vision-augmented trajectory prediction guided by textual descriptions," in *ECCV*. Springer, 2024, pp. 361–379.
- [59] N. Mu, J. Ji, Z. Yang, N. Harada, H. Tang, K. Chen, C. R. Qi, R. Ge, K. Goel, Z. Yang, S. Ettinger, R. Al-Rfou, D. Anguelov, and Y. Zhou, "Most: Multi-modality scene tokenization for motion prediction," in *CVPR*. IEEE, 2024, pp. 14 988–14 999.
- [60] K. Nakamura, R. Tian, and A. Bajcsy, "Not all errors are made equal: A regret metric for detecting system-level trajectory prediction failures," in *CoRL*. PMLR, 2024.
- [61] N. Nayakanti, R. Al-Rfou, A. Zhou, K. Goel, K. S. Refaat, and B. Sapp, "Wayformer: Motion forecasting via simple & efficient attention networks," in *ICRA*. IEEE, 2023, pp. 2980–2987.
- [62] J. Ngiam, V. Vasudevan, B. Caine, Z. Zhang, H.-T. L. Chiang, J. Ling, R. Roelofs, A. Bewley, C. Liu, A. Venugopal, G. S. Corrado, Q. V. Le, and R. Pang, "Scene transformer: A unified architecture for predicting multiple agent trajectories," in *ICLR*. OpenReview, 2022.
- [63] Z. Pang, D. Ramanan, M. Li, and Y.-X. Wang, "Streaming motion forecasting for autonomous driving," in *IROS*. IEEE, 2023, pp. 7407–7414.
- [64] S. Papais, L. Wang, B. Cheong, and S. L. Waslander, "Foresight: Multi-view streaming joint object detection and trajectory forecasting," in *ICCV*. IEEE, 2025.
- [65] D. Park, J. Jeong, S.-H. Yoon, J. Jeong, and K.-J. Yoon, "T4p: Test-time training of trajectory prediction via masked autoencoder and actor-specific token memory," in *CVPR*. IEEE, 2024, pp. 15 065–15 076.
- [66] D. Park, M. Surana, P. Desai, A. Mehta, R. M. John, and K.-J. Yoon, "Generative active learning for long-tail trajectory prediction via controllable diffusion model," in *ICCV*. IEEE, 2025, pp. 27 839–27 850.
- [67] M. Pei, S. Shi, X. Chen, X. Liu, and S. Shen, "Foresight in motion: Reinforcing trajectory prediction with reward heuristics," in *ICCV*. IEEE, 2025, pp. 28 303–28 312.
- [68] M. Peng, X. Guo, X. Chen, K. Chen, M. Zhu, L. Chen, and F.-Y. Wang, "LC-LLM: Explainable lane-change intention and trajectory predictions with large language models," *Communications in Transportation Research*, vol. 5, p. 100170, 2025.
- [69] N. Peri, J. Luiten, M. Li, A. Oşep, L. Leal-Taixé, and D. Ramanan, "Forecasting from lidar via future object detection," in *CVPR*. IEEE, 2022, pp. 17 181–17 190.
- [70] M. Pourkeshavarz, J. Zhang, and A. Rasouli, "Cadet: A causal disentanglement approach for robust trajectory prediction in autonomous driving," in *CVPR*. IEEE, 2024, pp. 14 874–14 884.
- [71] A. Prutsch, H. Bischof, and H. Possegger, "Efficient motion prediction: A lightweight & accurate trajectory prediction model with fast training and inference speed," in *IROS*. IEEE, 2024, pp. 9411–9417.
- [72] R. Qiu, J. Gong, X. Zhang, S. Luo, B. Zhang, and Y. Cen, "Adapting to observation length of trajectory prediction via contrastive learning," in *CVPR*. IEEE, 2025, pp. 1645–1654.
- [73] B. Rao, H. Liao, Y. Guan, C. Wang, B. Wang, J. Zhang, and Z. Li, "Amd: Adaptive momentum and decoupled contrastive learning framework for robust long-tail trajectory prediction," in *ICCV*. IEEE, 2025, pp. 28 849–28 858.
- [74] L. Rowe, M. Ethier, E.-H. Dykhne, and K. Czarniecki, "Fjimp: Factorized joint multi-agent motion prediction over learned directed acyclic interaction graphs," in *CVPR*. IEEE, 2023, pp. 13 745–13 755.
- [75] W. Ruan, X. Dang, Z. Zhou, S. Lyu, and Y. Liang, "RAST: A retrieval augmented spatio-temporal framework for traffic prediction," in *AAAI*. AAAI Press, 2026.
- [76] A. Seff, B. Cera, D. Chen, M. Ng, A. Zhou, N. Nayakanti, K. S. Refaat, R. Al-Rfou, and B. Sapp, "Motionlm: Multi-agent motion forecasting as language modeling," in *ICCV*. IEEE, 2023, pp. 8545–8556.
- [77] S. Shi, L. Jiang, D. Dai, and B. Schiele, "Motion transformer with global intention localization and local movement refinement," in *NeurIPS*, vol. 35. Curran Associates, Inc., 2022, pp. 6531–6543.
- [78] —, "MTR++: Multi-agent motion prediction with symmetric scene modeling and guided intention querying," *TPAMI*, vol. 46, no. 5, pp. 3955–3971, 2024.
- [79] N. Song, B. Zhang, X. Zhu, and L. Zhang, "Motion forecasting in continuous driving," in *NeurIPS*, vol. 37. Curran Associates, Inc., 2024, pp. 78 147–78 168.
- [80] Z. Su, C. Wang, D. Bradley, C. Vallespi-Gonzalez, C. Wellington, and N. Djuric, "Convolutions for spatial interaction modeling," in *CVPR*. IEEE, 2022, pp. 6573–6582.
- [81] J. Sun, X. Yue, J. Li, T. Shen, C. Yuan, S. Sun, S. Guo, Q. Zhou, and M. H. Ang Jr, "Impact: Behavioral intention-aware multimodal trajectory prediction with adaptive context trimming," *RA-L*, vol. 11, no. 1, pp. 610–617, 2026.
- [82] X. Tang, M. Kan, S. Shan, Z. Ji, J. Bai, and X. Chen, "Hpnet: Dynamic trajectory forecasting with historical prediction attention," in *CVPR*. IEEE, 2024, pp. 15 261–15 270.
- [83] S. Veer, A. Sharma, and M. Pavone, "Multi-predictor fusion: Combining learning-based and rule-based trajectory predictors," in *CoRL*, vol. 229. PMLR, 2023, pp. 2807–2820.
- [84] J. Wang, K. Messaoud, Y. Liu, J. Gall, and A. Alahi, "Forecast-PEFT: Parameter-efficient fine-tuning for pre-trained motion forecasting models," *CoRR*, vol. abs/2407.19564, 2024.
- [85] M. Wang, X. Zhu, C. Yu, W. Li, Y. Ma, R. Jin, X. Ren, D. Ren, M. Wang, and W. Yang, "Ganet: Goal area network for motion forecasting," in *ICRA*. IEEE, 2023, pp. 1609–1615.
- [86] X. Wang, T. Su, F. Da, and X. Yang, "Prophetnet: Efficient agent-centric motion forecasting with anchor-informed proposals," in *CVPR*. IEEE, 2023, pp. 21 995–22 003.
- [87] Y. Wang, C. Tang, L. Sun, S. Rossi, Y. Xie, C. Peng, T. Hannagan, S. Sabatini, N. Poerio, M. Tomizuka *et al.*, "Optimizing diffusion models for joint trajectory prediction and controllable generation," in *ECCV*. Springer, 2024, pp. 324–341.
- [88] Y. Wang, H. Zhou, Z. Zhang, C. Feng, H. Lin, C. Gao, Y. Tang, Z. Zhao, S. Zhang, J. Guo, X. Wang, Z. Xu, and C. Zhang, "Tenet: Transformer encoding network for effective temporal flow on motion prediction," *arXiv e-prints*, p. arXiv:2207.00170, 2022.
- [89] Z. Wang, Y. Wang, Z. Wu, H. Ma, Z. Li, H. Qiu, and J. Li, "Cmp: Cooperative motion prediction with multi-agent communication," *RA-L*, vol. 10, no. 4, pp. 3876–3883, 2025.
- [90] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou, "Chain-of-thought prompting elicits reasoning in large language models," in *NeurIPS*, vol. 35. Curran Associates, Inc., 2022, pp. 24 824–24 837.
- [91] D. Wen, Y. Wang, Z. Wu, Z. He, Z. Wu, and Z. Qingfang, "A driving-style-adaptive framework for vehicle trajectory prediction," in *NeurIPS*, vol. 37. Curran Associates, Inc., 2025, pp. 80 138–80 185.
- [92] J. Wiederer, J. Schmidt, U. Kressel, K. Dietmayer, and V. Belagianis, "Joint out-of-distribution detection and uncertainty estimation for trajectory prediction," in *IROS*. IEEE, 2023, pp. 5487–5494.
- [93] B. Wilson, W. Qi, T. Agarwal, J. Lambert, J. Singh, S. Khandelwal, B. Pan, R. Kumar, A. Hartnett, J. K. Pontes, D. Ramanan, P. Carr, and J. Hays, "Argoverse 2: Next generation datasets for self-driving

perception and forecasting,” in *NeurIPS*, vol. 36. Curran Associates, Inc., 2023.

[94] S. Woo, M. Kim, D. Kim, S. Jang, and S. Lee, “Fimp: Future interaction modeling for multi-agent motion prediction,” in *ICRA*. IEEE, 2024, pp. 14 457–14 463.

[95] W. Wu, X. Feng, Z. Gao, and Y. Kan, “Smart: Scalable multi-agent real-time motion generation via next-token prediction,” in *NeurIPS*, vol. 37. Curran Associates, Inc., 2024, pp. 114 048–114 071.

[96] L. Xiao, Z. Pan, Z. Wang, Z. Cao, and W. Li, “Srefiner: Soft-braid attention for multi-agent trajectory refinement,” in *ICCV*. IEEE, 2025, pp. 960–969.

[97] P. Xu, J.-B. Hayet, and I. Karamouzas, “Context-aware timewise vaes for real-time vehicle trajectory prediction,” *RA-L*, vol. 8, no. 9, pp. 5440–5447, 2024.

[98] R. Xu, X. Xia, J. Li, H. Li, S. Zhang, Z. Tu, Z. Meng, H. Xiang, X. Dong, R. Song, H. Yu, B. Zhou, and J. Ma, “V2v4real: A real-world large-scale dataset for vehicle-to-vehicle cooperative perception,” in *CVPR*. IEEE, 2023, pp. 13 712–13 722.

[99] Y. Xu, A. Bazarjani, H.-g. Chi, C. Choi, and Y. Fu, “Uncovering the missing pattern: Unified framework towards trajectory imputation and prediction,” in *CVPR*. IEEE, 2023, pp. 9632–9643.

[100] Y. Xu and Y. Fu, “Adapting to length shift: Flexilength network for trajectory prediction,” in *CVPR*. IEEE, 2024, pp. 15 226–15 237.

[101] Q. Yan, B. Zhang, Y. Zhang, D. Yang, J. White, D. Chen, J. Liu, L. Liu, B. Zhuang, S. Shi, and R. Liao, “Trajflow: Multi-modal motion prediction via flow matching,” in *IROS*. IEEE, 2025, pp. 3923–3928.

[102] Z. Yan, P. Li, Z. Fu, S. Xu, Y. Shi, X. Chen, Y. Zheng, Y. Li, T. Liu, C. Li, N. Luo, X. Gao, Y. Chen, Z. Wang, Y. Shi, P. Huang, Z. Han, J. Yuan, J. Gong, G. Zhou, H. Zhao, and H. Zhao, “Int2: Interactive trajectory prediction at intersections,” in *ICCV*. IEEE, 2023, pp. 8502–8513.

[103] Y. Yao, S. Yan, D. Goehring, W. Burgard, and J. Reichardt, “Improving out-of-distribution generalization of trajectory prediction for autonomous driving via polynomial representations,” in *IROS*. IEEE, 2024, pp. 488–495.

[104] M. Ye, J. Xu, X. Xu, T. Wang, T. Cao, and Q. Chen, “Bootstrap motion forecasting with self-consistent constraints,” in *ICCV*. IEEE, 2023, pp. 8470–8480.

[105] H. Yu, W. Yang, H. Ruan, Z. Yang, Y. Tang, X. Gao, X. Hao, Y. Shi, Y. Pan, N. Sun, J. Song, J. Yuan, P. Luo, and Z. Nie, “V2x-seq: A large-scale sequential dataset for vehicle-infrastructure cooperative perception and forecasting,” in *CVPR*. IEEE, 2023, pp. 5486–5495.

[106] Y. Yu, W. Han, L. Wu, B. Liu, E. Wang, and Z. Zhang, “Enduring, efficient and robust trajectory prediction attack in autonomous driving via optimization-driven multi-frame perturbation framework,” in *CVPR*. IEEE, 2025, pp. 17 229–17 238.

[107] W. Zhan, L. Sun, D. Wang, H. Shi, A. Clausse, M. Naumann, J. Kummerle, H. Konigshof, C. Stiller, A. de La Fortelle, and M. Tomizuka, “Interaction dataset: An international, adversarial and cooperative motion dataset in interactive driving scenarios with semantic maps,” *CoRR*, vol. abs/1910.03088, 2019.

[108] B. Zhang, N. Song, and L. Zhang, “Demo: Decoupling motion forecasting into directional intentions and dynamic states,” in *NeurIPS*, vol. 37. Curran Associates, Inc., 2024, pp. 106 582–106 606.

[109] L. Zhang, P. Li, S. Liu, and S. Shen, “Simpl: A simple and efficient multi-agent motion prediction baseline for autonomous driving,” *RA-L*, vol. 9, no. 5, pp. 4353–4360, 2024.

[110] Q. Zhang, S. Hu, J. Sun, Q. A. Chen, and Z. M. Mao, “On adversarial robustness of trajectory prediction for autonomous vehicles,” in *CVPR*. IEEE, 2022, pp. 15 138–15 147.

[111] Z. Zhang, A. Liniger, D. Dai, F. Yu, and L. Van Gool, “Trafficbots: Towards world models for autonomous driving simulation and motion prediction,” in *ICRA*. IEEE, 2023, pp. 1522–1529.

[112] Z. Zhang, A. Liniger, C. Sakaridis, F. Yu, and L. V. Gool, “Real-time motion prediction via heterogeneous polyline transformer with relative pose encoding,” in *NeurIPS*, vol. 36. Curran Associates, Inc., 2023, pp. 57 481–57 499.

[113] L. Zheng, S. Son, J. Liang, X. Wang, B. Clipp, and M. C. Lin, “Deep stochastic kinematic models for probabilistic motion forecasting in traffic,” in *IROS*. IEEE, 2024, pp. 943–950.

[114] X. Zheng, L. Wu, Z. Yan, Y. Tang, H. Zhao, C. Zhong, B. Chen, and J. Gong, “Large language models powered context-aware motion prediction in autonomous driving,” in *IROS*. IEEE, 2024, pp. 980–985.

[115] Y. Zhong, Z. Ni, S. Chen, and U. Neumann, “Aware of the history: Trajectory forecasting with the local behavior data,” in *ECCV*. Springer, 2022, pp. 393–409.

[116] Y. Zhou, H. Shao, L. Wang, S. L. Waslander, H. Li, and Y. Liu, “Smartrefine: A scenario-adaptive refinement framework for efficient motion prediction,” in *CVPR*. IEEE, 2024, pp. 15 281–15 290.

[117] Z. Zhou, J. Wang, Y. Li, and Y. Huang, “Query-centric trajectory prediction,” in *CVPR*. IEEE, 2023, pp. 17 863–17 873.

[118] Z. Zhou, Z. Wen, J. Wang, Y.-H. Li, and Y.-K. Huang, “QCNeXt: A next-generation framework for joint multi-agent trajectory prediction,” *CoRR*, vol. abs/2306.10508, 2023.

[119] Z. Zhou, L. Ye, J. Wang, K. Wu, and K. Lu, “Hivt: Hierarchical vector transformer for multi-agent motion prediction,” in *CVPR*. IEEE, 2022, pp. 8813–8823.

[120] Y. Zhu, D. Luan, and S. Shen, “Biff: Bi-level future fusion with polyline-based coordinate for interactive trajectory prediction,” in *ICCV*. IEEE, 2023, pp. 8226–8237.



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